MIL-D-45497 (Ord)
1 March 1960
SUPERSEDING
(See Section 6)

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M51
PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of detonator designated as Detonator, Electric, M51.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

UU-T-111 - Tape, Paper, Gummed (Sealing and Securing)
LLL-B-631 - Boxes, Fiber, Corrugated (For Domestic Shipment)

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.
MIL-STD-129 - Marking for Shipments and Storage.

DRAWINGS

ORDNANCE CORPS

8796020 - Detonator, Electric, M51, Assembly.

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PUBLICATIONS

ORDNANCE CORPS

ORD-P 1608-11 - Procedure and Tables for Continuous Sampling by Attributes.
OL-8802482 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 8796020, and with all requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be neat, clean and free from burrs sharp edges, excessive flash, unblended radii, distortion, surface defects, chips, dirt, grease, flux, rust, corrosion products (as applicable) and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The quality assurance provisions of this specification are applicable to inspection performed by or for the Government. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. Inspection shall be performed in accordance with this specification, Specification MIL-C-2550 and other specifications referenced in any of the contractual documents.

4.1.1 Contractor inspection.-Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

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4.1.2 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.3 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificates shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principal.

4.1.4 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied, and (b) Government product inspection. The

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Government inspector shall also ascertain prior to acceptance that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with.

4.1.4.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.4.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.1.4.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of parts produced by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting performance, safety or fit, as determined by the Government, shall not be cause for changing the lot number.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defect shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.3.

4.2.2.1 Plug and wire assembly (see dwg. 8796137 covering a detail of dwg. 8796070)

	Categories and defects	Method of inspection
Major	AQL 0.65 percent	
1.	Lead wire pull test	Test (see 4.3.1)
2.	Resistance between lead wires	Test (see 4.3.2)
3.	Distance between lead wires at plug	Gage
4.	Distance between bridge wires	Gage (see 6.2)
5.	Out of diameter of plug	Gage
6.	Length back to crimping surface	Gage
7.	Plug cracked or broken	Visual

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Minor AQL 2.50 percent

41. Depth to crimping surface.....Gage
42. Length of insulation, maximum (max.).....Gage
43. Length of lead wire, minimum (min.).....Gage
44. Wires not tinned.....Visual
45. Insulation damaged.....Visual
46. Angle missing.....Visual
47. Evidence of poor workmanship (see 3.3).....Visual

4.2.2.2 Cup (see dwg. 8796138 covering a detail of dwg. 8796020)

	Categories and defects	Method of inspection
Critical		
a.	Split, cut or crack through cup	Visual

Major AQL 0.65 percent

1. Inside diameter.....Gage
2. Total length.....Gage
3. Thickness through bottom.....Gage
4. Wall thickness.....Gage
5. Surface defects or distortion.....Visual
6. Foreign matter in interior.....Visual

Minor AQL 2.50 percent

41. Length of flare (when applicable).....Gage
42. Radius over max.....Gage
43. Surface finish improper.....Visual
44. Flare excessive (when applicable).....Visual
45. Evidence of poor workmanship (see 3.3).....Visual

4.2.2.3 Carton prior to sealing (see paragraphs 5.1.1.1 and 5.1.1.2)

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Incorrect number of rosetts	Visual (see 5.1.1.1)
2.	Incorrect number of cups	Visual (see 5.1.1.2)
Minor	AQL 0.40 percent	
41.	Wire insecure	Visual (see 5.1.1.1)

4.2.2.4 Sealed carton (see paragraph 5.4)

	Categories and defects	Method of inspection
Major	AQL 0.25 percent	
1.	Contents loose	Manual
2.	Tape incomplete or improperly assembled	Visual
Minor	AQL 1.50 percent	
41.	Carton damaged	Visual
42.	Flaps loose (when applicable)	Visual
43.	Marking misleadingoor unidentifiable	Visual

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4.2.3 Inspection equipment.-For the performance of the examinations and tests specified in 4.2.2 and 4.2.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment No. 8802482, or approved equivalents thereto. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Lead wire pull test.-The min. required force shall be applied axially between the plug and each lead wire.using equipment specified in 4.2.3. Observation shall be made to determine evidence of slippage of the lead wire within the plug.

4.3.2 Resistance between lead wires.-A direct current of 325 volts + 25 volts shall be applied to the lead wires and the resistance measured using equipment specified in 4.2.3. Observation shall be made for conformance to the drawing requirement.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C.-The detonator parts shall be packed for shipment as described below. The construction of the cartons and the material requirements of the carton and partitions shall comply with Federal Specification LLL-B-631. When assembled, the corrugations of the cartons and partitions shall parallel the carton depth. The manufacturer's joint on the carton will be metal stapled or stitched. Spacing of staples or stitches shall not average over 1 1/2 inches apart, center to center. A manufacturing tolerance of plus or minus 1/16 inches is permissible. The parts manufacturer shall be furnished preproduction samples of all packing materials so that proper fit of packing materials and metal parts can be determined. Minor modifications of packing material dimensions to provide a proper fit of all parts, as determined to be necessary by examination of sample packs, are permissible. All container voids shall be filled with dunnage to insure a tight packing.

5.1.1.1 Plug and wire assembly (see dwg. 8796137).-Ten thousand (10,000) plug and wire assemblies shall be packed in 40 rosettes of 250 each and secured by wrapping one wire around the extended sheaf of 249 wires. The 40 rosettes shall be packed in bulk in a 275 lb. test, RSC Style "B" flute, corrugated fiberboard carton. The inside dimensions of the carton shall be 21 by 10.5 by 8 inches.

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5.1.1.2 Cup.-Ten thousand (10,000) cups shall be packed in bulk in a 275 lb. test, RSC Style "B" flute, corrugated fiberboard carton. The inside dimensions of the carton shall be 10 by 10 by 5 inches.

5.2 Percentage of extra parts packaging.-When feasible, for small and light weight parts, the entire packing lot quantity including the percentage of extra parts shall be packed in the parts container.

5.3 Marking.-Marking of the packing shall comply with Standard MIL-STD-129.

5.4 Sealing.-The carton flaps shall be secured with gummed Kraft paper not less than 3 inches wide, complying with Federal Specification UU-T-111 for Class 2 and Class 3 tape. The tape shall be centered the full length of the outer flap seams, and extend over each end of the carton at least 3 inches. The flap edge seams shall be similarly sealed with the tape extending at least 3 inches along the corners of the box. Tape shall be properly moistened and sufficiently compressed to insure tight adhesion to the carton. Cartons weighing up to 50 lbs. shall be sealed with Class 2 tape. Cartons weighing 50 lbs. or more shall be sealed with Class 3 tape. Alternatively, when acceptable to the installation using the metal parts, flaps may be glued using a suitable adhesive applied over at least 75 percent of the contact area between inner and outer flaps. When flaps are pulled apart after gluing, they shall show complete fiber failure in the glued area.

5.5 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

5.6 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Kodak Contour Projector with surface illuminator, toolmakers microscope or an equivalent instrument may be used.

6.3 Supersession information.-Specification MIL-D-14172 (Ord) is superseded by Specification MIL-D-45497 (Ord) dated 1 March 1960 for the procurement of M51 Electric Detonator parts and MIL-D-14172A (Ord) dated 1 March 1960 for the loading, assembling and packing of same.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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